BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: 4471 / 4472

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q1. Q. 1

Q1. The capacitance of a charged capacitor is 'c' and energy stored on the account of charge is U then quantity of charge on the capacitor will be

(A) (B)

(C) (D)

Q2. If 4×10^{20} ev of energy is required to move a charge of 1C between two points then potential difference between the point is

(A) (B)

(C) (D)

Q3. A car battery has emf 12 v and internal resistance $5\times10^{-2}\ \Omega$, if it draws 60 A current , the terminal voltage of battery will be

(A) (B)

(C) (D)

Q4. The resistance of the coil of ammeter is R the shunt required to increase its range n times should have a resistance

(A) (B)

(C) (D)

Q5. When electron moves through a magnetic field, then change occurs in

(A) (B)

(C) (D)

Q6. The emf linked with same coil when the rate of change of current in the coil is unity is equal to

(A) (B)

(C) (D)

Q7. A coil having 500 square 100 ps, each of side 10 cm is placed normal to the magnetic field which is increasing at the rate of 0.1 tesla per second. The induced emf is

(A) (B)

(C) (D)

Q8. During frequency modulation when amplitude of signal is zero, the frequency of the carrier wave is

(A) (B)

(C) (D)

Q9. In RC series circuit voltage drops across resistor is 30 v and across capacitor is 40 v then the applied voltage must be

(A) (B)

(C) (D)

Q10. A Force F is needed to break a copper wire having radius R. The force needed to break a copper wire of same length and radius 2R will be

(A) (B)

(C) (D)

Q11. In common emitter transistor amplifier the input signal and output signal are always

(A) (B)

(C) (D)

Q12. The term inverter is used for

(A) (B)

(C) (D)

Q13. A proton and α-particle are accelerated through same voltages the ratio of their de-Broglie wavelength will be

(A) (B)

(C) (D)

Q14. When a photon collide with an electron which of the following of photon increases

(A) (B)

(C) (D)

Q15. The ratio of the longest and shortest wavelength of the Lyman series is approximately

(A) (B)

(C) (D)

Q16. The SI unit of radiation dose is

(A) (B)

(C) (D)

Q17. Leptons are particles that do not experience

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) What is RC time constant. Prove that unit of RC is second.

(ii) Define potential gradient. Give its direction and units.

(iii) If a point charge 'q' of mass 'm' is released in a non-uniform electric field with field lines pointing in the same direction, will it make a rectilinear motion?

(iv) Is E necessarily zero inside a charged rubber balloon if balloon is spherical? Assume that charge is distributed uniformly over the surface.

(v) Define current sensitivity of a galvanometer.

(vi) Why is there no work done by the magnetic force that acts on the charge.

(vii) If a charged particle moves in a straight line through some region of space, can you say that magnetic field in this region is zero?

(viii) Why the resistance of an ammeter should be very low.

(ix) What is the function of cadmium rods in a nuclear reactor.

(x) What is meant by dead time for G.M. counter. Give its value for G.M. tube.

(xi) How can radioactivity help in the treatment of cancer?

(xii) What is a radioactive tracer? Describe one application each in medicine, agriculture and industory.

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Is the filament resistance lower or higher in a 500 W, 220 V light bulb than in a 100 W, 220 V bulb?

(ii) In a R – L circuit , will the current Lag or Lead the voltage? Illustrate you answer by a vector diagram.

(iii) Draw a stress strain curve for ductile material and define yield point.

(iv) How does the motion of an electron in a n-type substance differ from the motion of holes in a p-type substance?

(v) What is drift velocity? Give its value.

(vi) What is the phenomenon of electroplating.

(vii) Give two uses of three phase A.C supply.

(viii) What is phase of A.C? How you express it by vector diagram.

(ix) Differentiate between unit cell and crystal lattice.

(x) Differentiate between elasticity and plasticity of a material.

(xi) What is normal operation of transistor? Show by diagram.

(xii) Can a transistor work as a switch? Explain.

Q4. (Marks 6 × 2 = 12) 4. Answer briefly any Six parts from the followings:-

(i) In a certain region the earth's magnetic field points vertically down. When a plane flies due north, Which wingtip is positively charged?

(ii) Does the induce emf always act to decrease the magnetic flux through a circuit?

(iii) Name the factors upon which the self inductance depends.

(iv) Does the brightness of a beam of light primarily depends on the frequency of photons or on the number of photons?

(v) What advantages an electron microscope has over an optical microscope?

(vi) Define stopping potential and threshold frequency in photoelectric effect.

(vii) Show that Compton shift is equal to Compton wavelength at an angle of 90°.

(viii) Is energy conserved when an atom emits a photon of light?

(ix) Differentiate between K_a X – rays and K_β X – rays.

Q. Section ---------- II

Note: Attempt any three questions. (8 × 3 = 24)

Q5. 5. (a) State Gauss's Law. Calculate the electric intensity due to an infinite sheet of charge.

(b) The potential difference between the terminals of a battery in open circuit is 2.2 V, when it is connected across a resistance of 5.0 Ω, the potential falls to 1.8 V. Calculate the current and the internal resistance of the battery.

Q6. 6. (a) What is transformer? Describe its principle, construction and working in detail.

(b) The resistance of a galvanometer is 50.0 Ω and reads full scale deflection with a current of 2.0 mA. Show by a diagram how to convert this galvanometer into voltmeter reading 200 V full scale

Q7. 7. (a) How can we use comparator as a Night Switch? Explain with the help of diagram.

(b) A 10 mH, 20 Ω coil is connected across 240 V and 180/π Hz source. How much power does it dissipate?

Q8. 8. (a) Write a note on Compton effect.

(b) A 1.0 m long copper wire is subjected to a stretching force and its length increases by 20 cm. Calculate the tensile strain and the percent elongation which the wire undergoes.

Q9. 9. (a) What is He – Ne Laser? Explain. Also write uses of laser in medicine and industry.

(b) A 75 Kg person receives a whole body radiation dose of 24 m-rad, delivered by α – particles for which RBE factor is 12. Calculate (i) absorbed energy in Joules (ii) Equivalent dose in rem

GROUP – II

Q1. Q. 1

Q1. If the distance between two point charge is halved, the electric intensity becomes

(A) (B)

(C) (D)

Q2. Current which flows from high potential to low potential is

(A) (B)

(C) (D)

Q3. The value of permeability of free space is

(A) (B)

(C) (D)

Q4. Lenz's law applies on

(A) (B)

(C) (D)

Q5. The mean value of A.C in a cycle is

(A) (B)

(C) (D)

Q6. Which one is a ductile substance.

(A) (B)

(C) (D)

Q7. Reverse current flows due to

(A) (B)

(C) (D)

> Note: A 'BLANK' cancellation stamp on the source scan partially covers Q7 option (C); it reads 'Electrons'.

Q8. Earth orbital speed is

(A) (B)

(C) (D)

> Note: A 'BLANK' cancellation stamp on the source scan partially covers Q8 option (C); it reads '30 km/s'.

Q9. Which of the series of hydrogen atom lies in ultraviolet region

(A) (B)

(C) (D)

> Note: A 'BLANK' cancellation stamp on the source scan partially covers Q9 option (C); it reads 'Paschen series'.

Q10. The binding energy per nucleon is maximum for

(A) (B)

(C) (D)

Q11. Which one is photo conductor

(A) (B)

(C) (D)

Q12. If the length and turns of a solenoid is doubled, strength of magnetic field will be

(A) (B)

(C) (D)

Q13. Energy stored in inductor is

(A) (B)

(C) (D)

Q14. In case of A.C through resistor, voltage and current are

(A) (B)

(C) (D)

Q15. A diode characteristic curve is plotted between

(A) (B)

(C) (D)

Q16. At low temperature, Body emitts radiation of

(A) (B)

(C) (D)

Q17. Which one is not affected by Electric and magnetic field.

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) How can you identify that which plate of a capacitor is positively charged?

(ii) Electric lines of force never cross. Why?

(iii) State Coulomb's law. Also write its mathematical form.

(iv) Write down at least two properties of electric field lines.

(v) Suppose that a charge 'q' is moving in a uniform magnetic field with a velocity v. Why is there no work done by the magnetic force that acts on the charge q?

(vi) Why does the picture on a T.V screen become distorted when a magnet is brought near the screen?

(vii) Define right hand rule for determining the direction of magnetic field in a current carrying conductor.

(viii) Find the value of the magnetic field that will cause a maximum force of 2.0×10^{-3} N on a 10 cm straight wire carrying a current of 5A.

(ix) Why are heavy nuclei unstable?

(x) What factor make a fusion reaction difficult to achieve.

(xi) Define mass defect and binding energy. Also write their mathematical expressions.

(xii) Show that 1u = 931 MeV by using the relation $E = mc^2$.

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Do bends in a wire effect its electrical resistance? Explain.

(ii) Why heat is produced in a conductor due to flow of current.

(iii) What are the difficulties in testing whether the filament of a lighted bulb obeys ohm's Law?

(iv) A sinusoidal current has rms value of 20 A. What is the maximum or peak value?

(v) How many times per minutes will be an incandescent Lamp reach maximum brilliance when connected to 50 Hz Source.

(vi) What is power factor in an A.C circuit? Explain.

(vii) Define stress and strain. What are their SI units?

(viii) Define modulus of elasticity. Show that the units of modulus of elasticity and stress are the same.

(ix) What is squids and where it is used?

(x) Define the current gain of a Transistor . Give Mathematical expression.

(xi) Why a photo diode is operated in reverse biased state?

(xii) How does the motion of an electron in a n-type substance differ from the motion of holes in p-type substance?

Q4. (Marks 6 × 2 = 12) 4. Answer briefly any Six parts from the followings:-

(i) In transformer, output power is less than input power. Why? Explain.

(ii) Can an electric motor be used to drive an electric generator with the output from the generator being used to operate the motor?

(iii) Does the induced emf always act to decrease the magnetic flux through a circuit?

(iv) What is frame of reference? Also differentiate between inertial and non-inertial frames of reference.

(v) Write the postulates upon which special theory of relativity is based.

(vi) Will higher frequency light eject greater number of electrons than low frequency light?

(vii) Is it possible to create a single electron from energy? Explain.

(viii) State two postulates of Bohr's model of the Hydrogen atom.

(ix) What are the advantages of lasers over ordinary light?

Q. Section ---------- II

Note: Attempt any three questions. (8 × 3 = 24)

Q5. 5. (a) Derive the expressions for electrical power and power dissipated in resistors.

(b) Two point charges, $q_1 = -1.0\times10^{-6}C$ and $q_2 = +4.0\times10^{-6}C$, are separated by a distance of 3.0 m. Find and justify the zero – field location.

Q6. 6. (a) Derive an expression of force acting on a moving charge in a magnetic field.

(b) An ideal step down transformer is connected to main supply of 240 V. It is desired to operate a 12 V, 30 W lamp. Find the current in the primary and the transformation ratio.

Q7. 7. (a) What is transistor? How can we use it as an amplifier. Find an expression for gain of an amplifier.

(b) A circuit has an inductance of 1/π H and resistance of 2000 Ω. A 50 Hz A.C is supplied to it. Calculate the reactance and impedance offered by the circuit.

Q8. 8. (a) Define photoelectric effect? Discuss it when the intensity of incident light remain constant.

(b) A 1.5 cm diameter cylinder is subjected to a load of 2500 kg. Calculate the stress on the bar in mega pascals.

Q9. (a) Explain the term mass defect and binding energy with an example.

(b) Compute the shortest wavelength radiation in the Balmer series? What value of n must be used?